(1) - Define Natural Language Processing (NLP). Provide three real-world applications of NLP and explain how they impact society.

Natural Language Processing (NLP) is a branch of artificial intelligence that enables computers to understand, interpret, and manipulate human languages. It combines computational linguistics with machine learning and deep learning models to process and analyze text and speech data.

Real-World Applications & Impact:

- **Machine Translation (e.g., Google Translate)** • Breaks down language barriers, allowing for global communication and access to information in various languages.
- **Sentiment Analysis** • Used by companies to analyze customer feedback or social media trends, helping businesses understand public perception and improve services.
- **Chatbots & Virtual Assistants** • Automate customer support and enhance user experience by providing instant responses to common queries.

(2) - Explain the following terms and their significance in NLP

1. Tokenization
2. Stemming
3. Lemmatization

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1. Tokenization
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(4) - Write a Python script to perform the following tasks on the given text:

- Tokenize the text into words and sentences.
- Remove stopwords from the tokenized text.
- Perform stemming and lemmatization on the tokens.
- Calculate the word frequency distribution.
- Generate a word cloud to visualize the most frequently occurring words.

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(6) - Create a TextBlob named exercise-blob containing "This is a TextBlob"

```
from textblob import TextBlob
exercise_blob = TextBlob("This is a TextBlob")
print(exercise_blob)

tokenized_text = exercise_blob.words
print(tokenized_text)
```

```
This is a TextBlob
WordList(['This', 'is', 'a', 'TextBlob'])
```

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(8) - Write a Python script to perform the following tasks on the given text:

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(9) - Write a regular expression to find all the hashtags in the following text:

```
text = "Loving the #sunshine and #beach vibes! #summer #vacation #travel #adventure #nature #photography #instagram #selfie #ootd #fashion #beauty #fitness #foodie #wanderlust #sunset #sunset #sunset
```